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A regionally-linked, dynamic material flow modelling tool for rolled, extruded and cast aluminium products

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ABSTRACT

A global aluminium flow modelling tool, comprising nine trade linked regions, namely China, Europe, Japan, Middle East, North America, Other Asia, Other Producing Countries, South America and Rest of World, has been developed. The purpose of the Microsoft Excel-based tool is the quantification of regional stocks and flows of rolled, extruded and casting alloys across space and over time, giving the industry the ability to evaluate the potential to recycle aluminium scrap most efficiently. The International Aluminium Institute will update the tool annually and publish a visualisation of results at www.world-aluminium.org/statistics/massflow.

Based on primary metal production, semi-fabricated products shipment and trade data from 1950s to 2014, the tool calculates regional domestic scrap availability and metal demand of the same alloy group and differentiates new and old scrap in each group at a given point in time. An intuitive user interface allows for changes in data inputs to generate bespoke results. To solve the mass balance difference of the 'Mining and Refining' and 'Aluminium Production' processes, which occur in the tool due to conflicting data, the software STAN was used.

Modelling of year 2014 stocks and flows indicate that three-quarters of all the aluminium ever produced is still in productive use. Over 26 million tonnes of new and old scrap are supplied to cast houses world-wide annually, in the form of mixed & casting scrap (11 million tonnes), used beverage cans (3 million tonnes), other rolled scrap (6 million tonnes), extruded scrap (4 million tonnes) and other scrap (2 million tonnes).

1. Introduction: a decade of knowledge building

This paper describes the methodology, data sources and results of a tool for conducting a dynamic material flow analyses (MFA) as described by Brunner and Rechberger (2004) of aluminium. A dynamic MFA is able to quantify total stocks accumulated over time; as distinct from a static MFA, which quantifies flows and stock changes of materials or substance within a system boundary at a certain point in time.

RWTH Aachen (Rombach et al., 2001a,b, 2002; Rombach, 2002; Zapp et al., 2002, 2003) and Alcoa (Bruggink and Martchek, 2004) were among the first to perform MFA for aluminium. The aluminium sector benefited from these and more recent studies described below, and is now working for over a decade; developing, testing and publishing their own MFA work (Boin and Bertram, 2005).

Built on decades of industry data and expert knowledge, reviews of published literature and views of product lifetimes and end-of-life (EOL) product collection rates, the first global aluminium flow model (Bertram et al., 2009) was developed in 200×, initially to map current and future availability of aluminium scrap. At that time (as today), worldwide statistical data on scrap availability and aluminium recycling was not comprehensively collected and published. Collecting meaningful and comprehensive recycling statistics has proven very difficult for the aluminium industry because there are more than 1500 remelting and refining plants worldwide, many of them in small and medium size, which can switch from tolling (not included in statistics) to purchasing (included in statistics) arrangements and from scrap-based to primary-based production depending on market prices and product requirements at any time.

As the model was refined, its potential as a communication, education, lobbying and strategic planning tool was realised, as well as its

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Table 1
Countries covered in each region.

Region	Countries	Comments
China	Mainland China	Important ingot producer and a consumer of final products.
Europe	EU28, Iceland, Macedonia, Moldavia, Norway, Serbia-Montenegro, Switzerland, Turkey, Ukraine	Covering regional Europe not political to make sure political changes do not affect the European model in the future.
Japan	Japan	Excellent data quality would otherwise be diluted in "Other Asia". High per capita consumer of final products.
North America	Canada, Mexico, USA	Important ingot producer and a consumer of final products. Mexico recycles a lot of scrap from the USA and ships secondary foundry castings back to the USA. Forming a unity.
Middle East	Bahrain, Oman, Qatar, Saudi Arabia, UA Emirates, Iran	Important primary ingot producer. Dynamic region. High per capita consumer of final products.
Other Asia	India, Indonesia, Malaysia, Pakistan, Singapore, South Korea, Taiwan, Thailand, Vietnam	Diverse group with poor data quality. Covering all bauxite, alumina and aluminium producers in Asia (excluding Japan and China)
Other Producing Countries	Australia, Azerbaijan, Cameroon, Egypt, Fiji, Guinea, Kazakhstan, New Zealand, Russia, Sierra Leone, South Africa, Tanzania, Zimbabwe	Large bauxite, alumina and primary producing region.
South America	Argentina, Brazil, Dominican Republic, Guyana, Haiti, Jamaica, Suriname, Venezuela	Covering bauxite mines and high income countries in this South America.
Rest of World	All other countries not listed above	No bauxite, alumina or primary aluminium producers included in this region. Importer of semis and final products.

power, when combined with lifecycle data, to model current and future environmental impacts (Martchek, 2006; International Aluminium Institute, 2009, 2016, 2017a).

Since 2006, the International Aluminium Institute (IAI) has published annual updates to this global model, as a Microsoft Excel workbook (International Aluminium Institute, 2015). This global model has played a major role in shaping a common understanding of the magnitude of aluminium stocks and flows from mining to use and recycling, was key in the development for common definition of recycling, and brought and still brings together aluminium producers, recyclers, semis produces and users of aluminium from all over the world giving them a platform to speak to each other.

In 2012, researchers at Vienna University of Technology (TU Wien) began to analyse aluminium material flows at country-level for Austria (Buchner et al., 2014, 2015a,b) and the aluminium content in buildings stock (Kleemann et al., 2014). In 2008, the Norwegian University of Science and Technology (NTNU) started a project to analyse the global aluminium cycle in more detail. This research entailed three dimensions: (i) the differentiation of alloys and alloying elements associated with aluminium, in order to anticipate and evaluate mitigation options related to potential quality challenges (Modaresi and Müller, 2012; Modaresi et al., 2014; Løvik et al., 2014; Løvic and Müller, 2014); (ii) the use of dynamic stock-driven aluminium flow models and scenarios to anticipate changes in energy use and greenhouse gas emissions related to aluminium production and to evaluate energy saving and climate change mitigation strategies (Liu et al., 2011, 2013); and (iii) a regional differentiation of the global aluminium cycle, tracking aluminium through 66 individual countries and regions, including the trade of aluminium contained in 126 product categories along the supply chain (Liu and Müller, 2013).

With time, users of the global IAI model had expressed a desire to understand not only the volumes and timing of scrap availability but also its location and quality. Requests were also made for regional differentiation in product lifetimes, power mixes and life cycle inventory data, in order to develop environmental impact scenarios.

Therefore, in 2012, IAI began to collaborate with NTNU to transform the global model into a number of discrete but interlinked regional models, drawing on its network of bauxite, alumina and aluminium producing member companies, manufacturing and mining experts, strategists and industrial ecologists. Since 2016 the results of these regional models (China, Europe, Japan, North America, Middle East, Other Asia, Other Producing Countries, South America and Rest of World) have been published as a visualisation of the global aluminium flow (International Aluminium Institute, 2017b).

A number of other research groups have also recently explored the aluminium cycle as part of focused projects: Yale University on US country level stocks and flows (Chen and Graedel, 2012); the University of Tokyo on aluminium and its alloying elements (Hatayama et al., 2007); Tsinghua University on flows in China (Chen et al., 2010) and the University of Cambridge Wellmet 2050 project MFA (Cullen and Allwood, 2013), based in part on the IAI global model. While these research programmes illustrate stocks and flows at a given moment in time, this research project is an ongoing effort to update regularly the aluminium MFA methodology, data input and output results.

The convergence of research activities of IAI, NTNU and TU Wien has resulted in a partnership between the three institutions, with the objective to publish a modelling tool that incorporates accurate production and demand data, trade data along the value chain, alloy group information and state of the art modelling expertise. The complete tool incorporates the regional models, a separate trade balancing tool and an interface, through which input data and assumptions can be manipulated and from which reports can be generated.

This paper describes the methodology, data quality and sources for the modelling tool, as well as output results for year 2014.

2. Methodology

2.1. Scope and system boundaries

The system described in this MFA is defined by both spatial and temporal boundaries (Chen et al., 2010). The spatial boundaries of the nine regional models is shown in Table 1, while the temporal boundary is the period 1950–2014. The starting year was chosen due to the availability of data from this point forward (World Bureau of Metal Statistics began collecting statistics on the aluminium industry in 1950) and to reflect the fact that the end of World War 2 marked a significant acceleration in aluminium production and demand; inclusion of earlier years would have meant inclusion of patchy and poor quality data and volumes so low as to not impact later stocks (and flows) significantly.

As shown in Fig. 1, the aluminium cycle is described in the regional models as a chain of six processes: bauxite mining & alumina production [mining & refining]; primary and recycled aluminium production [aluminium production]; semi-fabricated product manufacture [fabrication], production of final products [manufacturing], stock of products in use [use]; and management of end-of-life (EOL) products and new scrap [scrap recovery and trading].

Demand in the fabrication stage is met via aluminium production from one or more of the following metal sources: primary aluminium

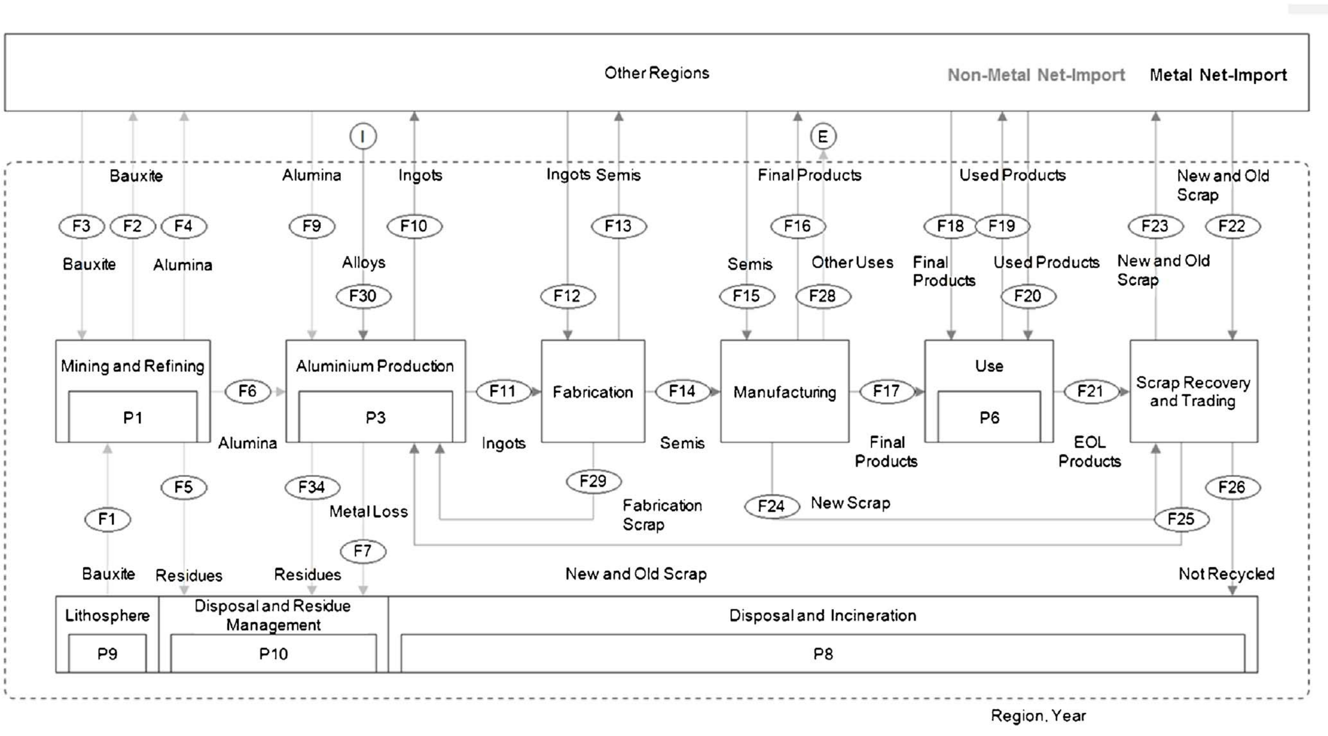


Fig. 1. Life cycle stages of aluminium.

Life cycle stages from mining to use and recycling. Light grey lines represent non-metal flows (bauxite, alumina and associated residues, metal loss) and dark grey the metal flows. Alloys (I) enter the aluminium cycle, whereas aluminium contained in powder and paste (E) exit the aluminium cycle.

production from bauxite ore (via mining & refining), recycled aluminium production from new and old scrap (via scrap recovery and trading) and internal remelting of fabrication scrap (via fabrication).

All flows are shown in aluminium mass equivalent values. For bauxite and alumina flows, where aluminium ions are bound in chemical compound form, the mass of aluminium is calculated based on the ratio of aluminium to other constituents, as defined in (Nunez and Jones, 2016). From semi-fabricated products to scrap, the flow of aluminium is split into four alloyed form subgroups (rolling, extrusion, casting and other) which are then further split into twelve product categories by market segment (outlined in Table 3).

As well as the flows between stages, the models include 5 stocks (see Fig. 1): stocks of bauxite ore and alumina at alumina refineries, ports and other storage facilities (P1); aluminium ingot stocks at production facilities, warehouses and in other storage (P3); the stock of final products in use (P6); metal in landfills (P8), the stock of chemically-bound aluminium in bauxite residue wastes from alumina refining processes and metal losses from primary and recycling metal production (P10). For the lithosphere (P9) only the stock change is investigated, to analyse the total stock an economic analysis of bauxite resources is needed which is outside the scope of this paper.

Inter-regional trade is accounted for at all life cycle stages. Through import and export each region is linked to one or more of the other eight regions.

Losses of bauxite and alumina are not the focus of the current models and are therefore only roughly estimated, while losses of metallic aluminium are analysed in detail.

3. Modelling

For ‘mining and refining’ and ‘aluminium production’ stages, where all flows are quantified independently, the total material input minus net stock change should be equal to the total output flows, otherwise a mass balance difference is introduced. To balance these processes, the software subSTance flow Analysis (STAN, 2016), developed by TU Wien, was used. The methodology followed is described in the “Handling of Inconsistent Data” section of this paper.

Aluminium ingot demand for the fabrication and the final product output of the manufacturing stages are not currently available via statistical data and are therefore calculated by a mass balance approach based on semis shipments for 12 end-uses (Building and Construction, Auto and Light Truck, Aerospace, Other Transport, Cans, Other Packaging-Foil, Machinery and Equipment, Cable, Other Electrical Products, Consumer Durables, Destructive Uses, and Other), utilisation rates (e.g., tonnage of windows produced per tonnage of extrusion input) and trade. Details on data quality, availability of statistical data and choices for modelling of data are outlined in the “Data Sources” section of this paper.

Apart from packaging applications, the timing of EOL products exiting the use stage is calculated using a normal distribution, with a coefficient of variation equal to 1/3 of the mean product lifetime. While there are more adequate distribution curves (such as the Gamma or Weilbul functions), which avoid the potential for negative values, it was decided to use a normal distribution because it is familiar to non-MFA specialists and therefore intuitive to users of the tool. In addition, the probability that a normal random variable takes a negative value is about 0.135%, negligibly small from a practical point of view. Packaging is assumed to have a one-year lifetime (the model defines product lifetimes in whole year units only), i.e. discarded in the year following its production. The modelling tool allows the user to assign different lifetimes to different alloy groups within one final product segment, e.g. a window (extrusion) in a building might have a different lifetime than a door handle (casting) or cladding (rolling). Due to lack of data this possibility is not executed in the results shown below. Chen and Shi (2012) note that the choice of product lifetime distribution does not impact the results of a dynamic MFA, whereas the choice of mean lifetime has a significance impact.

The mass of aluminium recycled from products post-use is calculated as a function of collection rate (how many EOL products are collected for recycling), pre-melting process recovery rate (e.g., shredders, used sorting technology) and melting recovery rates (melting, dross recycling and salt slag recovery). After collection and pre-melting recovery the material is termed “old scrap”. For scrap generated at the fabrication and manufacturing stage a 100% collection and pre-melting rate is assumed. Scrap generated at the fabrication stage is termed “fabrication scrap” while scrap generated at the manufacturing stage is termed “new scrap”. In reality, fabrication, new and old scrap are almost impossible to differentiate, particularly when they have passed through traders’ hands. In this tool, new scrap is traded together with old scrap, while fabrication scrap is not on the market for trading and therefore recycled in the same region as it is generated.

The post-use fate of machinery and equipment, electrical products and consumer durables is not known therefore the not-recycled proportion of this mass could still be in use as a second hand product or landfilled. In this paper this not-recycled aluminium is allocated to use stage.

Inter-regional trade is mapped between all stages of the aluminium cycle; the methodology is illustrated and described in detail in Liu and Müller (2013). Each bulk trade flow (physical movement of product x, from region A to region B) is represented in reported trade statistics (United Nations, 2016) by two data points: import by region B from region A (reported import) and export from region A to region B (reported export). In theory these reported data should be equal, but in practice they rarely have the same magnitude. To overcome this issue and to minimise uncertainty in results generated by the model using trade data, a set of “confidence” rules were used to choose between import and export data for a given flow, where the two did not equate (see Table 2). This table was used for all products except alumina exported from Other Producing Countries to China, Other Asia, South America and Rest of World. This is because the alumina export trade data for Australia (the largest exporter in the Other Producing Countries region) is not split up by country, to protect company confidentiality. Trade data for the Middle East should be treated with some caution since Iran does not report to the UN. North America, Europe and Japan are assumed to have a better reporting system and are therefore having the higher confidence level than China, Middle East, Other Asia, Other Producing Countries, South America and Rest of World. If trade is between these regions, then import data are chosen over export data, since it is assumed that more rigorous checking of the imported goods occurs due to the levelling of tariffs on imports.

Once each bulk product trade flow (e.g. number of cars) has been chosen, it is multiplied by an assumed average aluminium content (e.g. 120 kg Al per car), based on data prepared by Liu and Müller (2013). For final products, this calculated aluminium trade flow is split into the four alloy forms. It is assumed that the distribution of exported alloy forms in a given product category is the same as the distribution of production in the exporting region, e.g. if the share of final products in the consumer durables category produced in region A is 50% rolled and 50% extruded, then the export of consumer durables from region A is assumed to comprise 50% rolled and 50% extruded products, too. Alloy group exports from all regions for one product type are summed at the global level and distributed in the same ratio to importing regions. For example, the total export of machinery from all regions is 1.8 million tonnes in 2014 with a share of 40% rolled products, 40% extrusions and 20% castings. Europe importing 0.3 million tonnes will then receive this quantity in the form of 40% rolled, 40% extruded and 20% cast products.

Table 2
“Confidence” rules for trade.

Importer/Exporter	CN	EU	JP	ME	NA	OA	OP	SA	ROW
CN		EU	JP	ME	NA	OA	OP	SA	ROW
EU	EU		JP	EU	NA	EU	EU	EU	EU
JP	JP	EU		JP	NA	JP	JP	JP	JP
ME	CN	EU	JP		NA	OA	OP	SA	ROW
NA	NA	EU	JP	NA		NA	NA	NA	NA
OA	CN	EU	JP	ME	NA		OP	SA	ROW
OP	CN	EU	JP	ME	NA	OA		SA	ROW
SA	CN	EU	JP	ME	NA	OA	OP		ROW
ROW	CN	EU	JP	ME	NA	OA	OP	SA	

CN: China, EU: Europe, JP: Japan, ME: Middle East, NA: North America, OA: Other Asia, OP: Other Producing Countries, SA: South America, ROW: Rest of World.
Importing regions are shown across the top row, exporting regions down the first column. For instance, a product traded from South Africa to China would use data reported by China, but a product traded from China to South America would use data reported by South America.

The methodology described above is used for all trade flows with the exception of used products. These products are not accounted for in the scrap trade statistics and therefore need to be modelled separately. In this case, the regions are categorised as one of two types: exporters (Europe, Japan, Middle East and North America) and importers (China, Other Asia, Other Producing Countries and South America). The region Rest of World is assumed to be neither an importer nor an exporter (in balance). It is assumed that exporting regions export 20% of their used vehicles and 20% of their used consumer durables (by mass of aluminium). Of this total exported volume, it is estimated that China imports 70% (by mass) and the other three importing regions share the remaining 30% equally. Imported used products are assumed to be entering the use stage to be re-used again and so their product lifetime is reset to year zero. They are assumed to enter the scrap recovery and trading stage at the end of their second in-use life.

The material flow calculations for China, Europe, Japan, North America, Other Asia, Other Producing Countries and South America are modelled on both statistical data (on semis shipments, primary production and trade flows) and on informed estimates, such as manufacturing utilisation rates, product lifetimes and scrap collection rates. For the Middle East and Rest of World, no robust dataset for semis shipments is available. Therefore, this regional model is based on primary production and net ingot imports data, with coefficients for semis shipment distribution and an iterative calculation.

4. Data sources

Annual statistical datasets for bauxite, alumina and primary aluminium production and semis shipments are used to populate the model for the years 1950–2014. Bauxite production data is from United States Geological Survey (2016); bauxite qualities, bauxite stocks CRU (2016a) and ingot stocks CRU (2016b) from CRU; primary aluminium production statistics for the years 1950–1972 from World Bureau of Metal Statistics (1984). All other data are aluminium industry data, collated by IAI and the national and regional aluminium associations listed in “Acknowledgments”. A sample data set for 2014 is shown in Table 3. Since semis shipments are in many regions considered confidential to the industry, only total shipments and not the split by rolling, extrusion, casting and others are shown.

Trade data are downloaded from the UN Commodity Trade Statistics website (United Nations, 2016) and include: bauxite (S1-2833); metallurgical grade alumina (S1-51365); aluminium ingots (S1-6841); semis (S1-68421, S1-68422, S1-68423, S1-68424, S1-68425, S1-68426); final products (115 flows, for further information refer to Liu and Müller, 2013); and scrap (S1-28404). Used products (second hand and/or EOL vehicles and used consumer durables) are based on informed estimates.

Default average product lifetimes, end of life collection rates and pre-melting processing rates are informed estimates. For more than 15 years an industry working group, which includes expertise in life cycle analysis, recycling, sustainability and packaging, has met on a regular basis to share published estimates and hard regional data and to compare methodologies to develop such informed estimates. In the MFA tool all estimates and hard data can be changed. Melting rates are based on Boin and Bertram (2005). Lifetimes are resolved at the ‘region-product category-alloy form’ level and are not changeable per year.

Table 3
Bauxite mining, alumina refining, primary aluminium production and semis shipments in 1000 tonnes, 2014.

	CN	EU	JP	ME	NA	OA	OP	SA	ROW
Bauxite Mining (bulk)	74,000	3900		1100		19,900	93,400	43,000	
Alumina Refining (bulk)	49,300	5700		200	5600	3800	24,200	13,200	
Primary Aluminium Production (Al)	28,300	4200		5200	4600	2100	8000	1500	
Semis Shipments (Al Alloy)	30,900	11,900	3400	*1800*	10,600	7700	3800	2300	*1400*
Bldg & Const	9200	2700	600	*600*	1400	2600	800	400	*400*
Transportation: Auto & Lt Truck	4400	3000	1500		2300	1200	500	200	*200*
Transportation: Aerospace	100	100			400				
Transportation: Truck/Bus/Trailer/Rail/Marine/Other	1900	1400	100	*300*	1000	1000	200	200	*100*
Packaging: Cans	600	1200	400	*200*	1900	400	300	600	*100*
Packaging: Other (Foil)	1800	1000	100	*100*	400	200	300	100	*100*
Machinery & Equipment	3300	700	200	*200*	800	300	300	100	*100*
Electrical: Cable	3100	500		*100*	400	1000	600	200	
Electrical: Other	1900	500	100	*200*	500	100	200		*100*
Consumer Durables	2300	400	100	*100*	700	300	100	200	*100*
Other (excl. Destructive Uses)	1000	200	200		500	300	300	100	*100*
Destructive Uses	1400	300	100		100	200	200	100	*100*

CN: China, EU: Europe, JP: Japan, ME: "Middle East", NA: "North America", OA: "Other Asia", OP: "Other Producing Countries", SA: "South America", ROW: "Rest of World". Semis shipments for "Middle East" and "Rest of World" (numbers in italic) are calculated.

Collection, processing and melting rates can also change over time [region-product category-alloy form-year level].

Of all the functions in the models, results are most sensitive to the informed estimates of product lifetime and EOL collection rate; therefore, a range of values should be explored when running sensitivity analyses. The figures shown in Table 4 are mean lifetimes per each product category (the same lifetime is used for each alloy form) and weighted average rates (based on EOL products available in 2014) for collection, processing and melting. Due to the high value of aluminium scrap from fabrication and manufacturing processes, collection rate and pre-melting rates are set to 100% in all regions.

Most ingots are shipped in alloyed form, rather than pure aluminium and alloying elements need to be added as an input to the production stage. The mass of alloying elements added to primary aluminium is based on data published by European Aluminium (2000) and for recycled aluminium, information published by Boin and Bertram (2005). Coefficients on metal losses during casting processes (primary and recycling route) are also based on research by Boin and Bertram (2005). As outlined above, bauxite mining and alumina refining are not the focus of this analysis; losses of non-metallic aluminium are based on Liu and M & ller (2013).

5. Handling of inconsitent data

In the developed material flow modelling tool, the uncertainty of data and estimates is not considered. Furthermore, mass balance stocks are introduced at the "Mining and Refining" and the "Aluminium Production" stages, where input data and output data are independently computed and are not equal (see Fig. 2a).

In reality, each data input used in this analysis is associated with an uncertainty. For example, human error can occur during the production of statistics or collection rates of EOL products are sometimes based on informed estimates, on single case studies or on complete statistics. Laner et al. (2016) have published a well-defined approach to ensure a consistent definition of data uncertainty in quantitative form.

For this analysis, the authors inserted the 2014 flows (e.g.,

Table 4
Default average lifetimes and weighted average recycling rates, 2014.

	CN	EU	JP	ME	NA	OA	OP	SA	ROW
Lifetimes									
Building and Construction	35	60	60	50	60	50	50	50	50
Auto & Lt Truck	25	15	12	12	15	30	30	30	30
Aerospace	40	40	40	40	40	40	40	40	40
Truck/Bus/Trailer/Rail/Marine/Other	30	30	30	30	30	30	30	30	30
Packaging	1	1	1	1	1	1	1	1	1
Machinery & Equipment	40	40	40	40	40	40	40	40	40
Electrical – Cable	40	40	40	40	40	40	40	40	40
Electrical – Other	20	20	20	20	30	20	20	20	20
Consumer Durables	10	8	8	8	8	15	15	15	15
Other (ex Destructive Uses)	20	20	20	20	20	20	20	20	20
End-of-life collection rates	76%	81%	89%	72%	76%	69%	78%	85%	70%
Old scrap pre-melting processing rates	95%	98%	98%	98%	98%	95%	98%	98%	98%
Old scrap melting rates	95%	97%	97%	97%	95%	95%	95%	96%	95%
New scrap melting rates	96%	98%	98%	98%	96%	96%	96%	96%	96%
Fabrication scrap melting rates	97%	99%	99%	99%	97%	97%	97%	97%	97%

Lifetimes are based on informed estimates; recycling rates are calculated as the weighted average of the recycling rates of 40 products (10 main products split into 4 alloy groups). These product rates are based on statistics (e.g., collection rate of cans in North America), studies (e.g., collection rate for aluminium building products in Europe – European Aluminium, 2004) and informed estimates (e.g., collection rate of end-of-life vehicles in Other Asia).

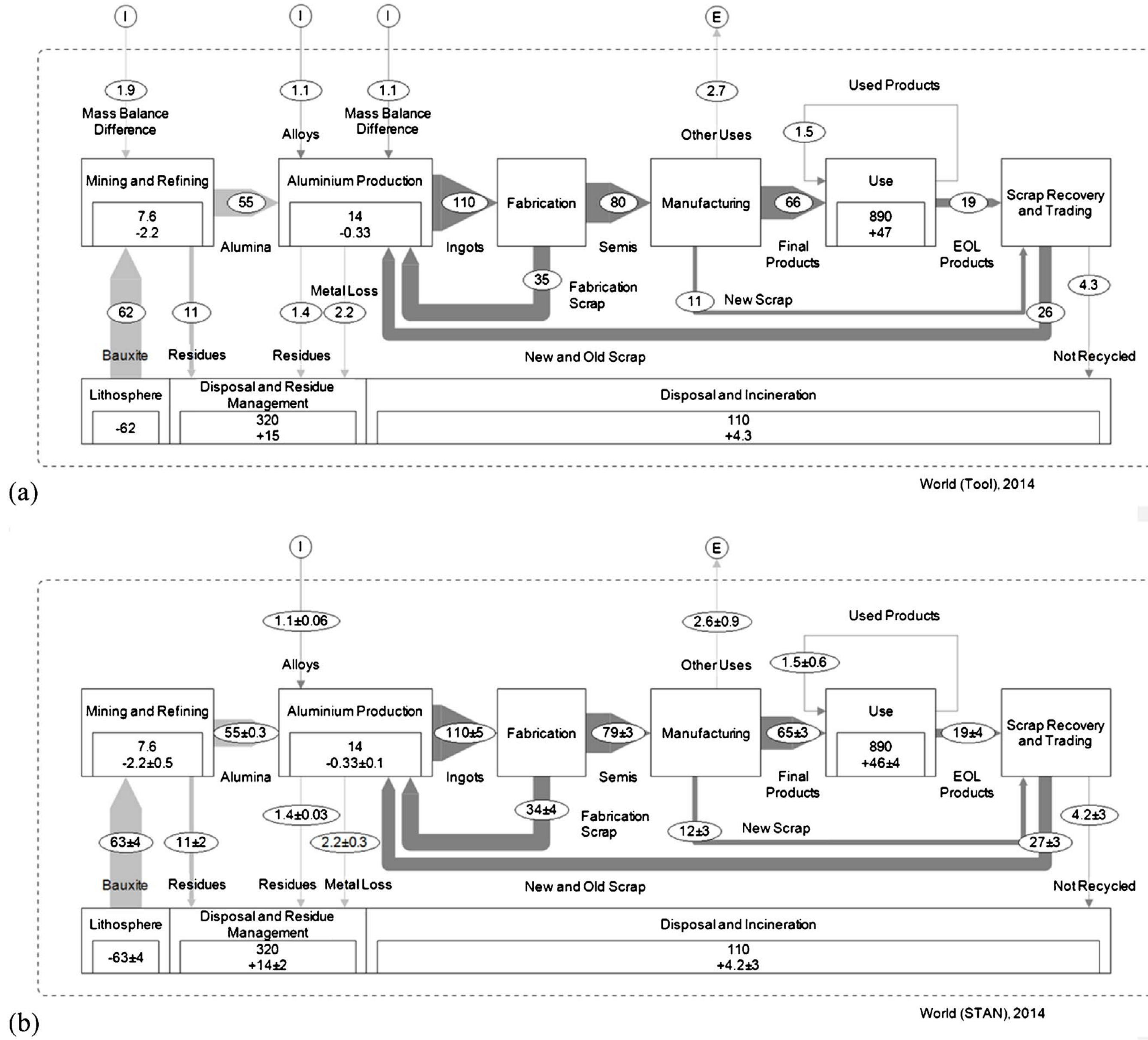


Fig. 2. 2014 Global aluminium cycle: (a) without uncertainty calculation (Tool) and (b) uncertainty calculation (STAN).
Values are in million tonnes. Stocks and flows are rounded to two significant digits, while uncertainties are rounded to one significant digit. Process in- and outputs might not add up due to rounding. (I) and (E) flows are used to describe mass balance differences in 2a (inflows, stock change and outflows of a process are defined independently and do not equal zero) as well as alloys entering the system or other uses exiting the aluminium cycle in 2a and b. Light grey lines represent non-metal flows and dark grey the metal flows. All flows are normalised to their aluminium content (bauxite, alumina and associated residues, metal loss). The total stock in 2014 equals the stock in 2013 plus the stock change in 2014 (e.g., total use stage 2014 in World (STAN) = 890 million tonnes + 46 million tonnes = 940 million tonnes). The confidence interval of uncertainties is 68%. For example, for "new and old scrap", we can be 68% confident that between 24 million and 30 million tonnes are entering recycling production worldwide.

aluminium ingot import to China from Other Asia), stock changes (e.g., ingot stocks in 2014 minus ingot stocks in 2013) and transfer coefficients (e.g., collection rate of end-of-life vehicles in Europe) into the software STAN (2016), together with the defined uncertainties based on Laner et al. (2016) for all data input, which can be found in the supporting material (Appendix A). Herewith, STAN calculated the most probable result for all stock changes and flows, as well as the uncertainties. To sum up uncertainties (e.g., uncertainties of the 8 aluminium ingot import flows and the 8 aluminium ingot export flows of North America to one uncertainty for aluminium net imports to North America) error propagation was used. Results of this analysis are shown in Fig. 2b. Historical flows and stock changes are not part of this analysis since STAN is not able to undertake dynamic modelling in the use stage, therefore is not able to calculate the uncertainty of the stocks.

6. Aluminium stocks and flows

Results of running the regional models with default assumptions and using STAN are presented in Fig. 3. Regional flow models without uncertainty calculations can be found in the supporting material

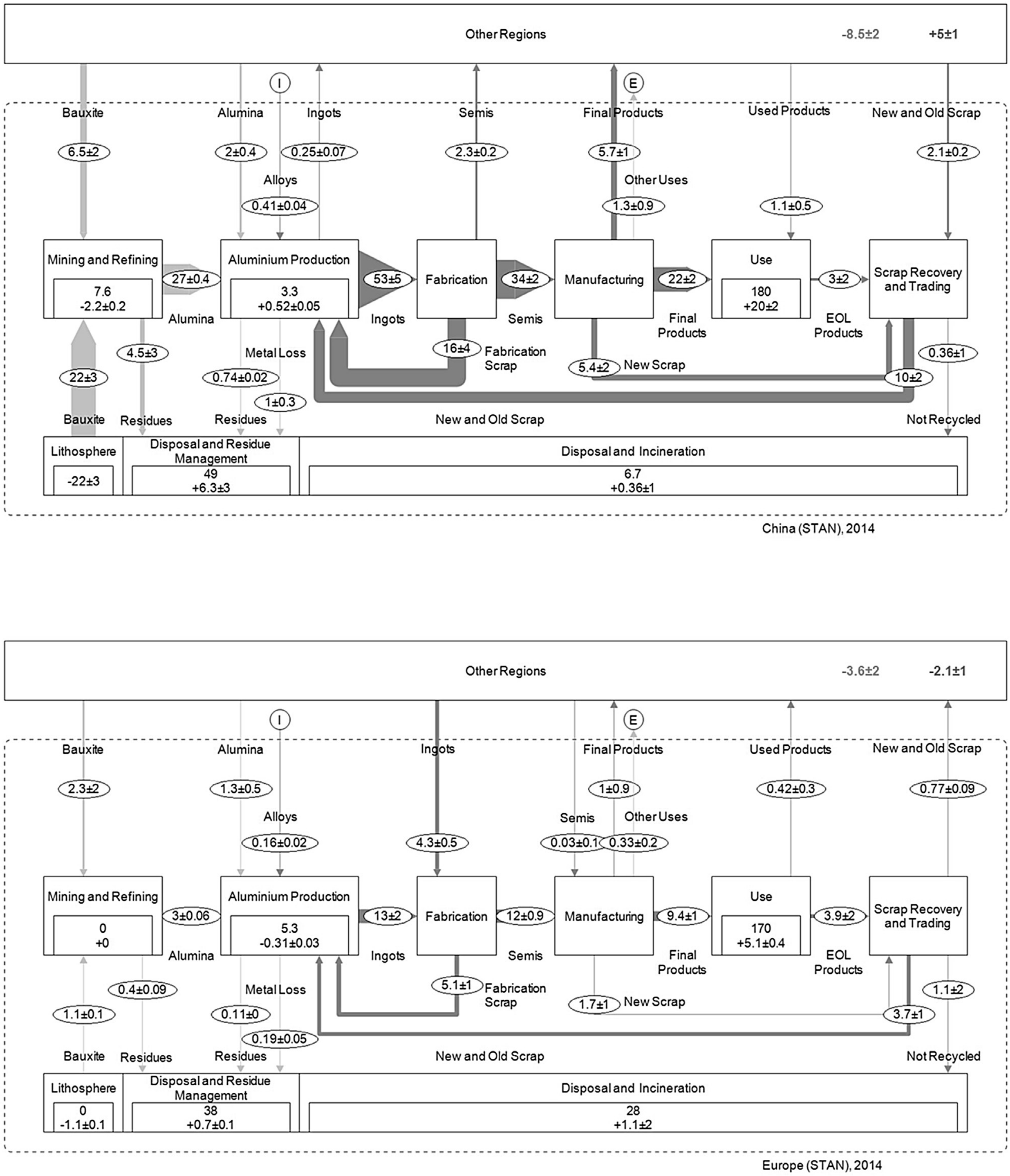
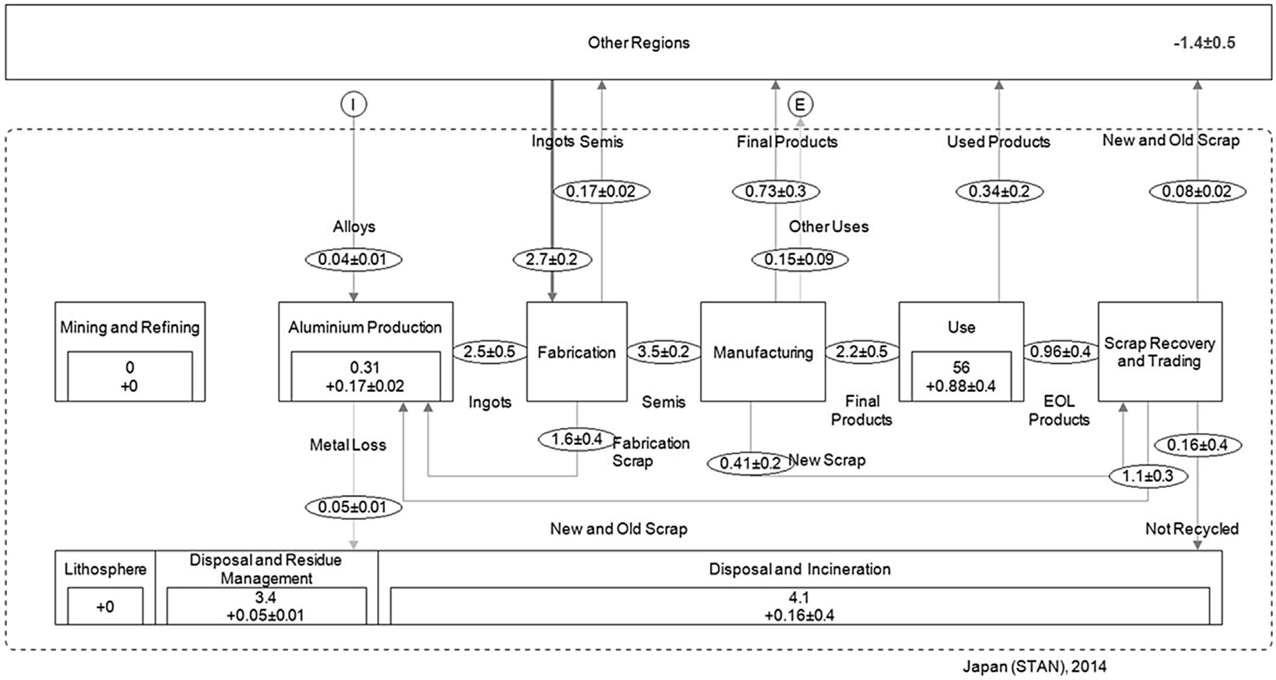


Fig. 3. 2014 regional aluminium cycles with uncertainty calculation (STAN).
See Fig. 2 for figure captions.

(Appendix B). It should be emphasised that these models have been developed for use by members of the IAI and regional and national aluminium associations, as well as by individuals and institutions with an interest in MFA. Default data and assumptions are therefore changeable and can be changed based on the specific knowledge or assumptions of the user. The results presented here are based on consensus assumptions and represent the best data available today.

In 2014, aluminium stock in use in China (180 million tonnes) is the



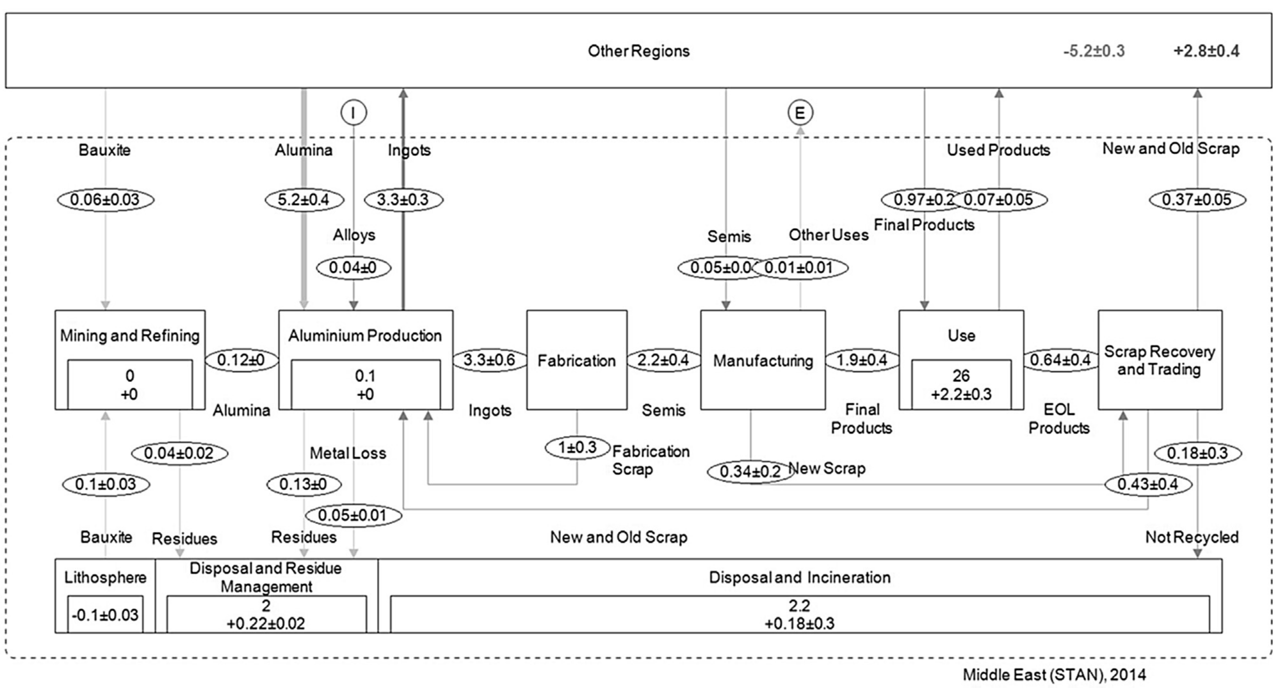


Fig. 3. (*continued*)

same magnitude as that of North America (210 million tonnes) and Europe (170 million tonnes). Nevertheless, the net annual increase in Chinese in use stocks is currently almost four times higher than that of Europe or North America. All three regions import bauxite and alumina to supply their smelting industries. In addition, Europe and North America are importing ingots to meet metal demand. China is a significant net exporter of semi-finished aluminium (2 million) and of aluminium in final products (6 million tonnes), and only in recent years has shown a net export of ingots (250,000 t). Both Europe (800,000 t) and North America (1.6 million tonnes) are net exporters of aluminium

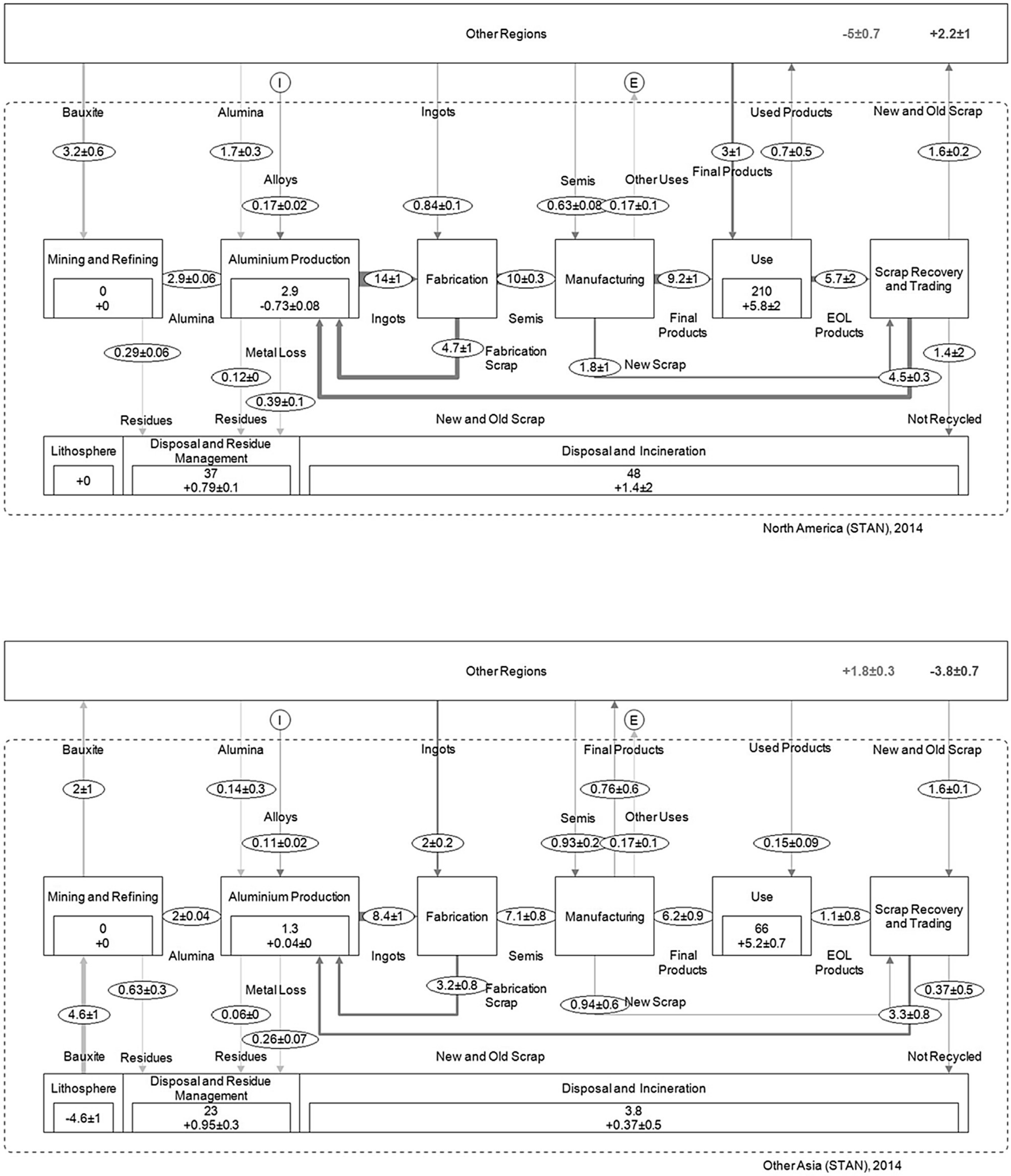


Fig. 3. (*continued*)

scrap, while China is a net importer of 2 million tonnes of scrap.

Other Asia shows a very strong presence at all stages of the cycle. It exports bauxite at the beginning of the cycle, has a significant net annual increase in stock in use, and is an importer of scrap. Another fast developing region is the Middle East, with a mature but growing primary aluminium industry, growth in fabrication, investments in a nascent mining and refining industry and increased demand for final products, particularly in building and construction.

South America and Other Producing Countries function mainly as bauxite, alumina and ingot suppliers for all other regions. Japan has the

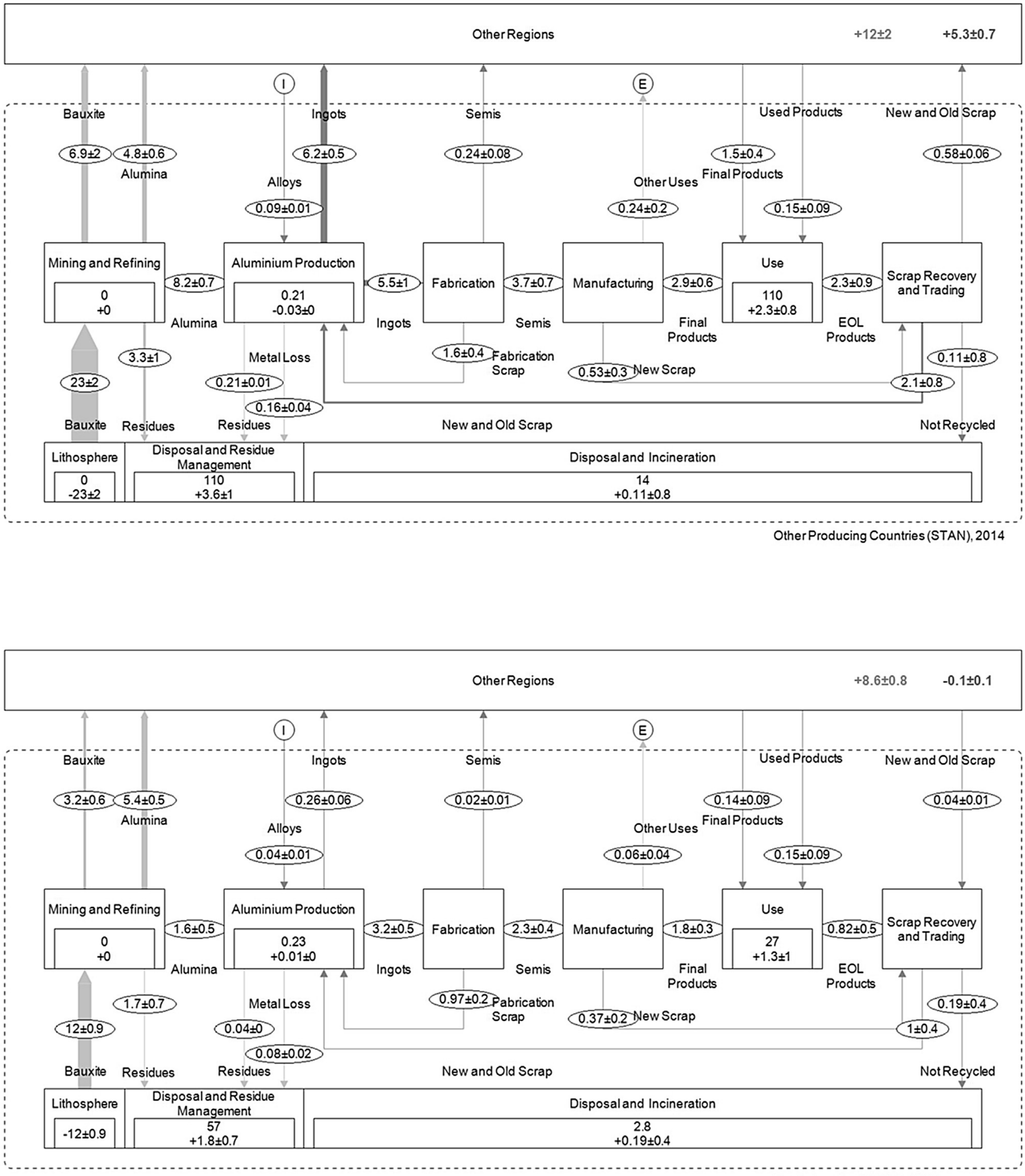


Fig. 3. (*continued*)

smallest net increase in stock in use and its aluminium production (2.5 million tonnes) is based entirely on domestic scrap. Semis demand requires that this scrap supply is supplemented by almost 3 million tonnes of imported ingots.

A cumulative 1.1 billion tonnes of primary aluminium were produced between 1950 and 2014 (another 20,000 t between 1888 and 1949), of which 860 million tonnes are still in use and a further 80 million tonnes non-recycled products (products where the post-use fate

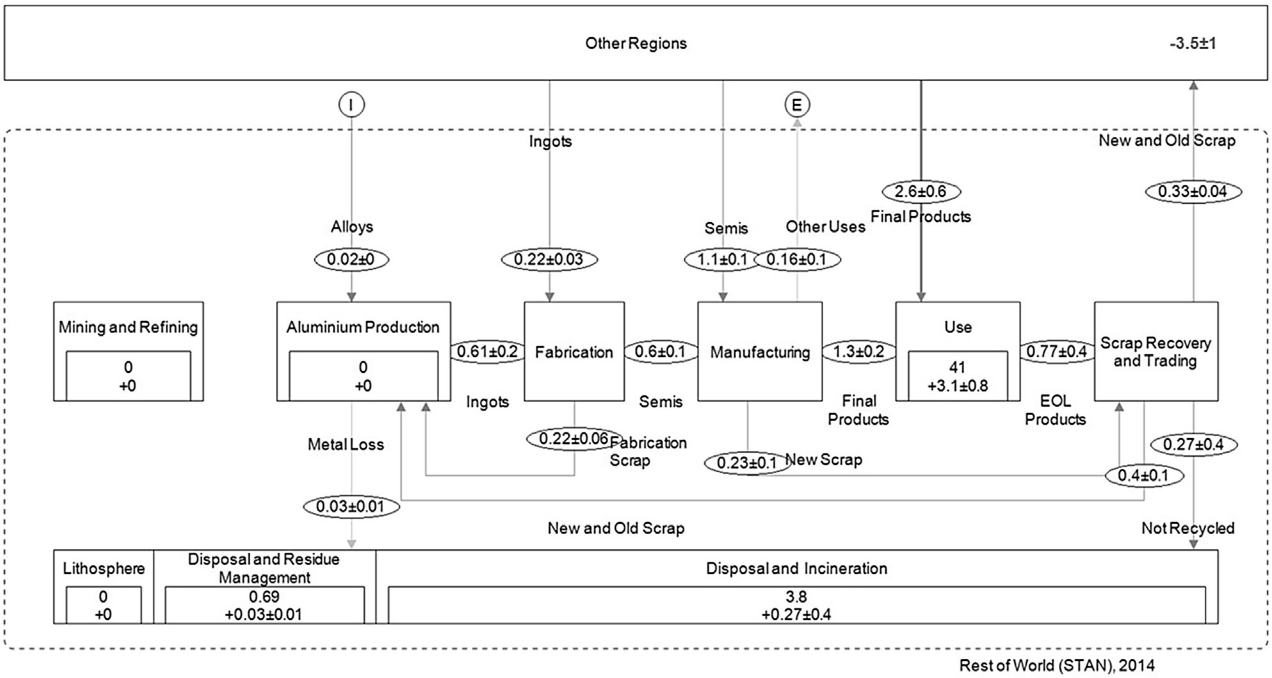


Fig. 3. (*continued*)

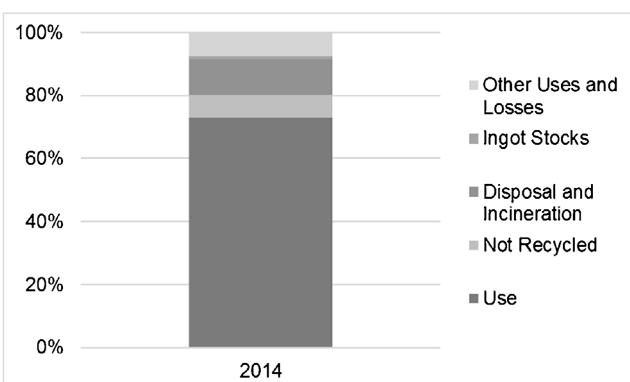


Fig. 4. Destination of primary aluminium ever produced since 1888.

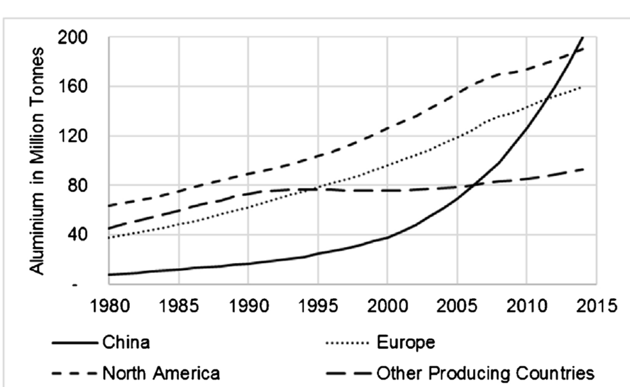


Fig. 5. Total regional annual aluminium stock in use in selected regions from 1980 to 2014.

is not known) could still be in use (which gives a total of 940 as shown in Fig. 2). The remaining 200 million tonnes is landfilled or has otherwise exited the cycle (as metal loss or as uses where the metallic properties are lost, such as energy recovery – incineration). Hence around 75% of all aluminium ever produced is still in productive use (Fig. 4). As shown in Fig. 5, the majority of the 860 million tonnes is located in four regions, China (ca. 20%), North America (ca. 20%), Europe (ca. 20%) and Other Producing Countries (ca. 10%). The accumulation of these stocks show exponential growth for China, a linear increase for North America and Europe from 1980 to 2014 and no growth for Other Producing Countries between 1990 to 2005 with a slight increase of 1–2% annually from 2005 to 2014.

7. Regional scrap availability and demand by alloy group

Over 26 million tonnes of new and old scrap are supplied to cast houses world-wide annually, in the form of mixed & casting scrap (11 million tonnes), used beverage cans (3 million tonnes), other rolled scrap (6 million tonnes), extruded scrap (4 million tonnes) and other scrap (2 million tonnes). Fig. 6 illustrates the actual (scrap mixing) and theoretical (alloy to alloy sorting) supply-demand balance for recycled casting alloys in 2014, globally and for selected regions. The supply and demand balance for rolled and extruded alloys can be found in Appendix C and 4 respectably.

As shown in Fig. 6g and h there was, worldwide in 2014, a recycled casting ingot demand of 14 million tonnes. Should all scrap be sorted 100% by alloy type (6h) then this global demand would be filled by 6 million tonnes of recycled castings, leaving an additional demand of 8 million tonnes. Due to the fact that sorting technologies cannot achieve 100% and that what is currently technologically feasible today is not employed by all, rolled and extruded scrap is being mixed with casting scrap. Therefore, the actual amount of recycled castings produced is 11 million tonnes (6g). This leaves a total additional recycled casting ingot demand of 3 million tonnes. For North America (6e) and Europe (6c) the amount of recycled casting ingots that could be produced from

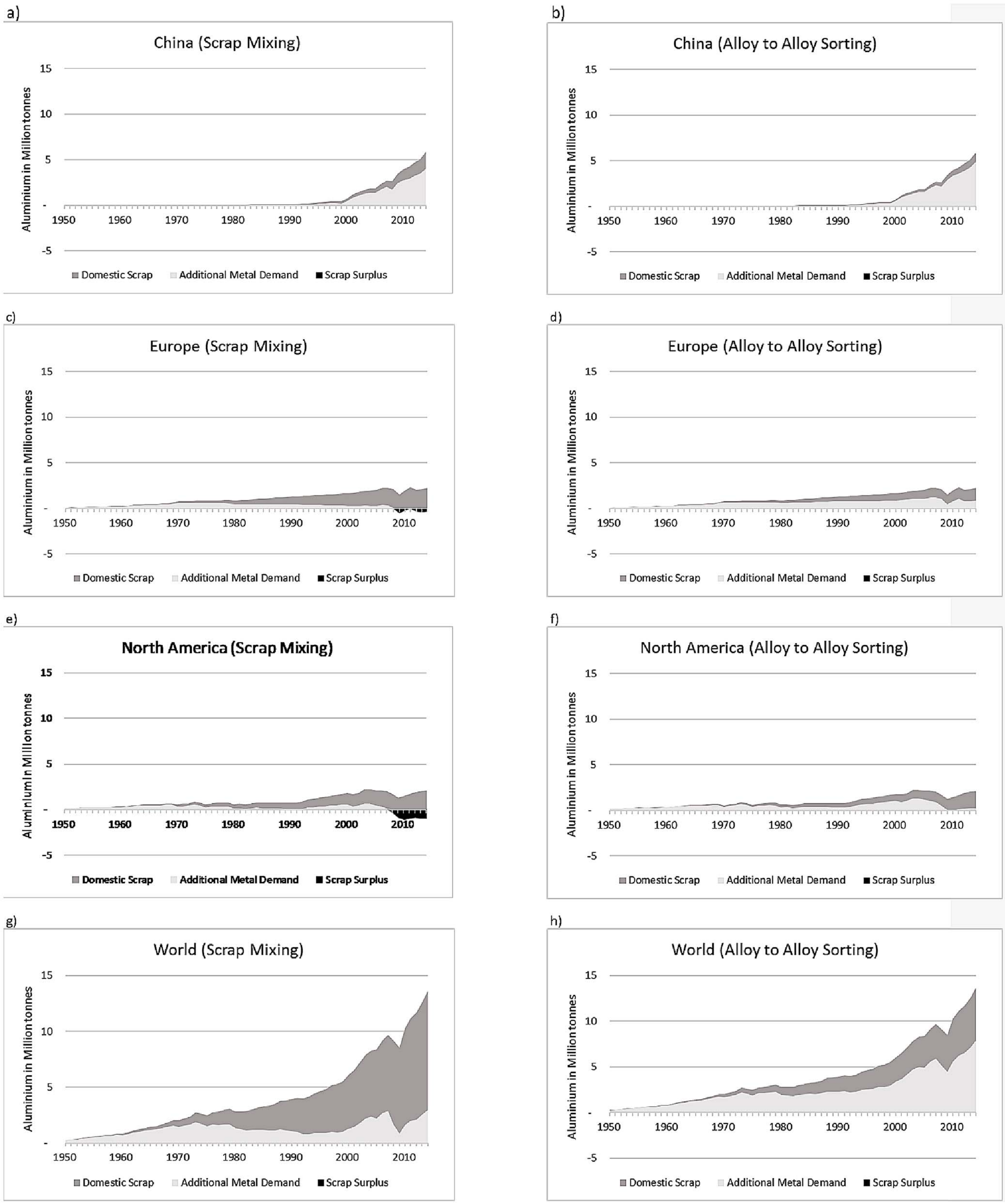


Fig. 6. Recycled casting ingot demand and supply balance – globally and in selected regions.

The total area under the curve represents recycled casting ingot demand. It is made up in part by recycled casting ingots that could be produced from domestically available scrap (dark grey). If demand is higher than theoretical production, then a region needs to import metal in the form of scrap or ingots (light grey). If the demand is lower than the theoretical production, then a region needs to export scrap (black). Low alloy sorting (scrap mixing) results in a casting scrap surplus for Europe (c) and North America (e) (black). In China (a) and globally (g) a casting scrap deficit currently exists (light grey). Where scrap to be sorted alloy to alloy then all regions would experience a casting scrap deficit.

Table 5
Comparison of key results in this paper with published literature.

	China				Europe		North America				Middle East		South America	
	A08	B08	A09	C09	A08	B08	A08	B08[a]	A09	D09[a]	A08	B08	A08	B08
Stock in Use	90	70	100	80	100	100	200	200	200	200	20	20	20	30
Net Addition to Stock in Use	10	8	10	10	6	6	5	3	3	0.4	2	2	1	2
New and Old Scrap Used	5	3	6	4[b]	4	5	3	4	3	3	0.3	0.4	1	0.7
Net Import	4	2	6	8	8	6	7	6	6	5	2	1	−10	−6
Not Recycled	0.4	1	0.4	1	1	3	2	3	2	2	0.1	0.5	0.1	0.9

Values in million tonnes, rounded to one significant digit.
A08: This paper, 2008 data. A09: This paper, 2009 data. B08: Lui and Müller (2013), 2008 data. C09: Chen and Shi (2012), 2009 data. D09: Chen and Graedel (2012), 2009 data.
[a] USA only.
[b] Old Scrap only.

domestic scrap is larger than the demand for recycled castings by 1 million and 400,000 t respectably. This is reflected by the fact that both regions are net exporters of unsorted scrap. If North America were to recycle all its scrap domestically today, then alloy to alloy separation for wrought alloys (rolling and extrusion scrap) would need to increase from 70% to 95% (25% increase). For Europe, an increase from 65% to 75% (10%) would be needed. If China does not want to put primary metal or more rolling and extrusion scrap into recycled castings (6a), then it is in need of 4 million tonnes additional recycled casting ingots. This again corresponds well with the fact that China was a net importer of unsorted scrap in 2014. Should Europe (6d), North America (6f) and China (6b) sort 100% by alloy, then China's demand for recycled castings would increase further, Europe's oversupply would turn into a shortfall and North America would have a balance between demand and supply.

8. Comparison with other published studies

Løvik and Müller (2014) and Modaresi et al. (2014) simulated future production of 7 alloy groups for vehicles (one end-use), and the relative share of demand met from primary and recycled sources. In this paper 12 end-uses and 3 alloy groups (rolled, extruded and cast alloys) are investigated. Therefore, comparisons of results are difficult. Liu and Müller (2013) developed country and regional aluminium flow charts globally, Chen and Shi (2012) for China and Chen and Graedel (2012) for the USA. As shown in Table 5, key results of this paper for the years 2008 and 2009 are compared with these publications. Most data are of the same magnitude, with two exceptions. Chen and Graedel (2012) concluded a 'net addition to stock in use' of 400,000 t for the USA, compared to 3 million tonnes for North America in this paper. This is mainly due to the significant lower inflow of aluminium into the use stage used by Chen and Graedel (2012) for the year of the financial crisis. Secondly, the amount of 'not recycled aluminium' for China is 400,000 t in this paper, compared to 1 million tonnes in Chen and Shi (2012) and Liu and Müller, 2013Liu and Müller (2013). Here the reasons are manifold, although different lifetimes and end-of-life collection rate estimates are the most significant.

9. Conclusion and outlook

The regionally-linked dynamic material flow modelling tool is developed upon the state of the art methodology and best available data for aluminium cycle. It can be used to develop if-then-scenarios at global (e.g. scrap intake rate) and regional (e.g. old rolled scrap generation) levels. The user is able to parse default or self-generated data and assumptions and to run the models or even change trade flows and look at impacts. Some key parameters have been compared with previously published data and most are found to be of similar magnitude.

As for any dynamic MFA, results are very sensitive to product lifetime assumptions. Here it is very difficult to improve the results since no hard data exist and what is correct for one year might not be in the following. In the next version of this model a variable lifetime is suggested for building and construction segment products in China and Other Asia, since during an urbanisation phase lifetime could be assumed to be shorter than after saturation.

Another complex area is collection rate assumptions at end-of-life; these also have a great impact on the model results. Here the tool is continuously updated with new publications to represent the current knowledge. Another update to the tool, will be to better understand the various categories of aluminium process loss and the reactions causing such. The focus will be on the loss of aluminium when becoming liquid, during surface protection and during the use phase.

Acknowledgements

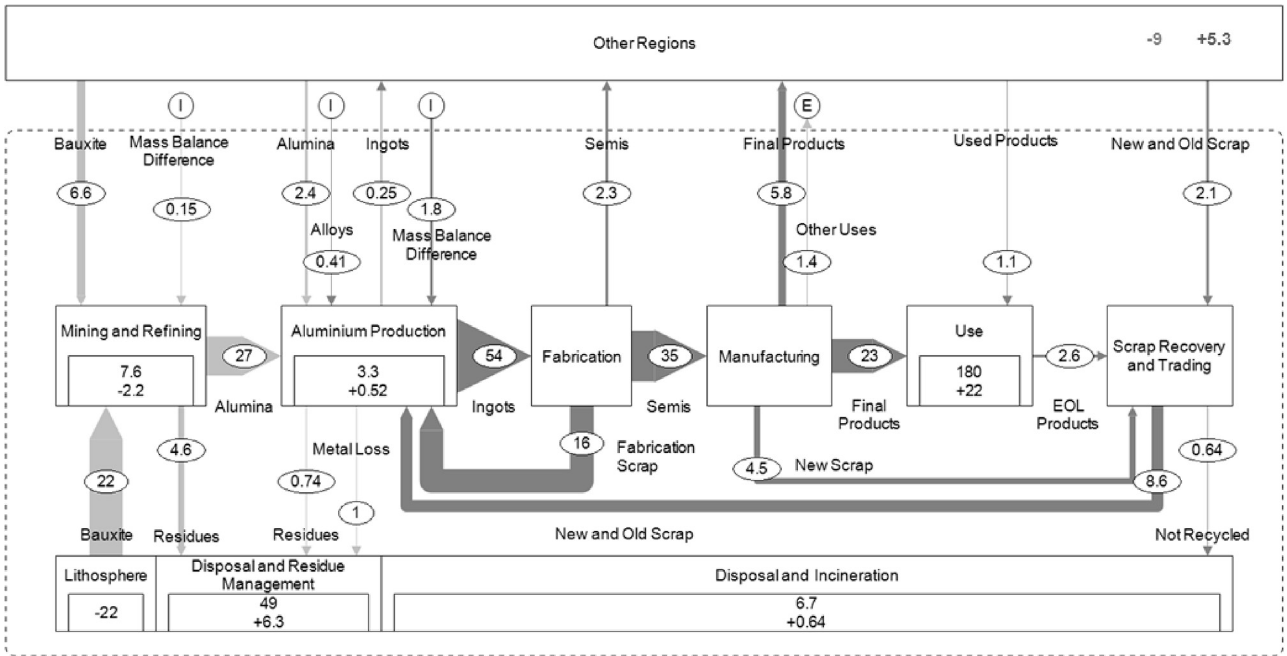
We would like to thank The Aluminum Association (USA), Australian Aluminium Council, Aluminium Federation of South Africa, Brazilian Aluminium Association, China Nonferrous Metals Industry Association, European Aluminium, Gesamtverband der Aluminiumindustrie e.V. (Germany) and Japan Aluminium Association for continuously supporting the Material Flow Project over the past 15 years. Without their data on semis shipments and regional expertise the high quality of the model would have not been possible. Furthermore, we thank Eoin Dinsmore and Marco Georgiou from CRU, who have helped significantly with industry data and expert knowledge with the regionalisation.

Appendix A. Uncertainties used in STAN

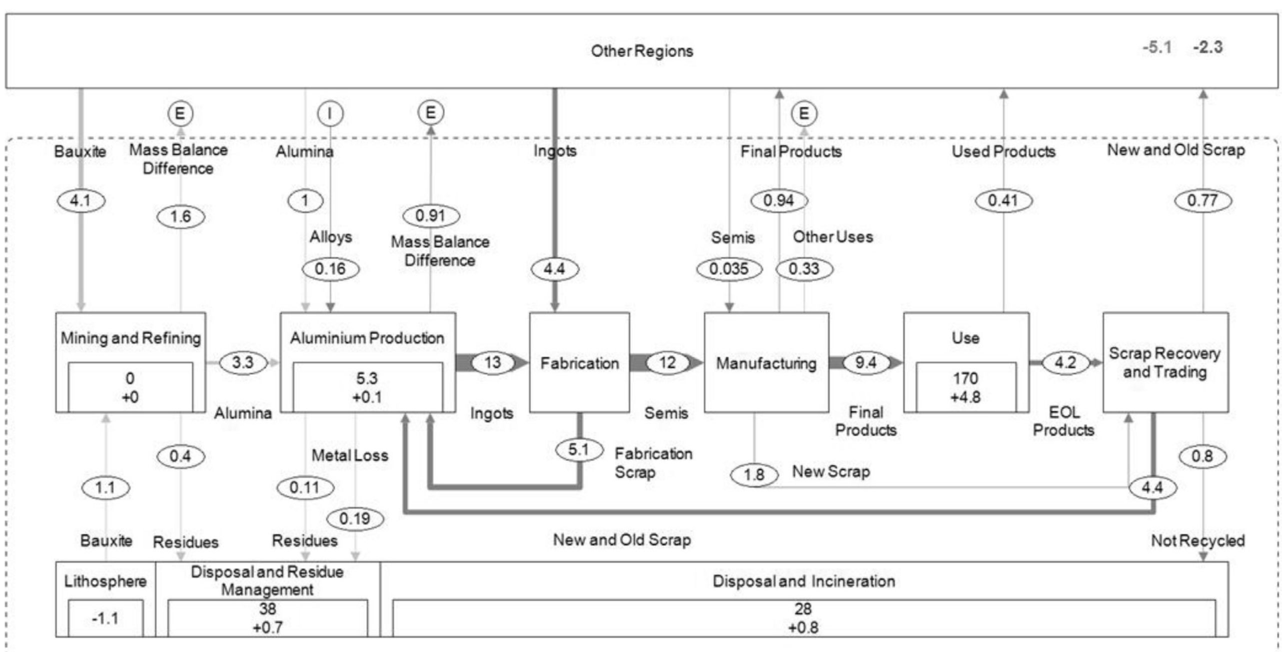
	Reliability	Completeness	Geographical Correlation	Temporal Correlation	Other	Total
Bauxite Trade	20.6%	0.0%	0.0%	0.0%	41.3%	46.2%
Mining China	62.3%	4.5%	0.0%	0.0%	41.3%	74.9%
Mining Other Regions	6.8%	0.0%	0.0%	0.0%	10.3%	12.3%
Mining Other Asia	62.3%	4.5%	0.0%	0.0%	41.3%	74.9%
Mining Other Producing Countries	20.6%	0.0%	0.0%	0.0%	13.7%	24.7%
Mining South America	20.6%	0.0%	0.0%	0.0%	41.3%	46.2%
Mining Middle East	62.3%	0.0%	0.0%	0.0%	20.3%	65.5%
Bauxite Demand	2.3%	0.0%	0.0%	0.0%	4.5%	5.1%
Ore Residue	62.3%	0.0%	2.3%	0.0%	20.6%	65.7%
Alumina Trade	6.8%	0.0%	0.0%	0.0%	41.3%	41.9%
Refining	2.3%	0.0%	0.0%	0.0%	0.0%	2.3%
Aluminium Trade	6.8%	0.0%	0.0%	0.0%	13.7%	15.3%
Alumina Demand	2.3%	0.0%	0.0%	0.0%	0.0%	2.3%
Bauxite Residue	20.6%	0.0%	6.8%	0.0%	0.0%	21.7%
Smelting	2.3%	0.0%	0.0%	0.0%	0.0%	2.3%
Alumium Shipments	20.6%	0.0%	0.0%	0.0%	13.7%	24.7%
Semis Trade	6.8%	4.5%	0.0%	0.0%	13.7%	15.9%
Semis Shipments						0.0%
China	6.8%	0.0%	0.0%	0.0%	0.0%	6.8%
Europe	6.8%	4.5%	0.0%	0.0%	0.0%	8.2%
Japan	6.8%	0.0%	0.0%	0.0%	0.0%	6.8%
Middle East	62.3%	41.3%	4.5%	0.0%	0.0%	74.9%
North America	2.3%	0.0%	0.0%	0.0%	0.0%	2.3%
Other Asia	20.6%	13.7%	0.0%	0.0%	0.0%	24.7%
Other Producing Countries	20.6%	13.7%	0.0%	0.0%	0.0%	24.7%
South America	20.6%	13.7%	0.0%	0.0%	0.0%	24.7%
Other	62.3%	41.3%	4.5%	0.0%	0.0%	74.9%
Destructive Uses	62.3%	13.7%	0.0%	0.0%	0.0%	63.8%
EOL Products	62.3%	41.3%	0.0%	0.0%	0.0%	74.7%
Final Product Trade	6.8%	0.0%	0.0%	0.0%	41.3%	41.9%
Alloying Elements	6.8%	0.0%	10.3%	10.3%	0.0%	16.1%
Metal Loss	6.8%	0.0%	20.6%	20.6%	0.0%	29.9%
Scrap Trade	6.8%	0.0%	0.0%	0.0%	13.7%	15.3%
Spend Potlinings	2.3%	0.0%	2.3%	0.0%	0.0%	3.3%
Used Products	62.3%	41.3%	0.0%	0.0%	0.0%	74.7%
TC Fabrication	6.8%	0.0%	10.3%	10.3%	0.0%	16.1%
TC Manufacturing	13.7%	0.0%	41.3%	41.3%	0.0%	60.0%
TC Import of Used Products	20.6%	20.6%	0.0%	20.6%	20.6%	41.2%
TC Collection Rate	41.3%	0.0%	0.0%	0.0%	41.3%	58.4%
Stock Change Use - China	20.6%	0.0%	0.0%	0.0%	4.5%	21.1%
Stock Change Use - EU, NA, JP, SA	62.3%	0.0%	0.0%	0.0%	4.5%	62.5%
Stock Change - OT, OA, ME	20.6%	0.0%	0.0%	0.0%	13.7%	24.7%
Stock Change - OP, OT	62.3%	0.0%	0.0%	0.0%	4.5%	62.5%
Stock Change Ingot	6.8%	4.5%	0.0%	4.5%	4.5%	10.3%

Total = SQRT(Reliability^2 + Completeness^2 + Geographical Correlation^2 + Temporal Correlation^2 + Other^2)
White- highly sensitive, light grey – medium sensitive, dark grey – not sensitive.
For further information see Laner et al. (2016).

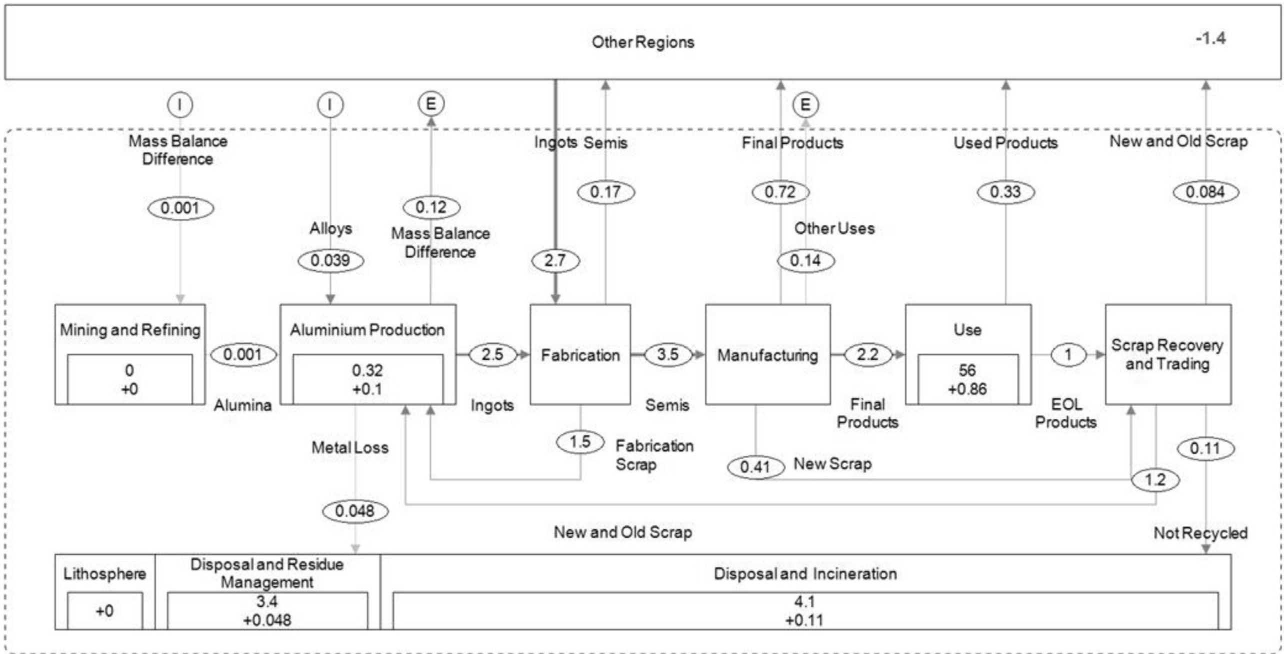
Appendix B. 2014 regional aluminium cycles without uncertainty calculation (Tool)



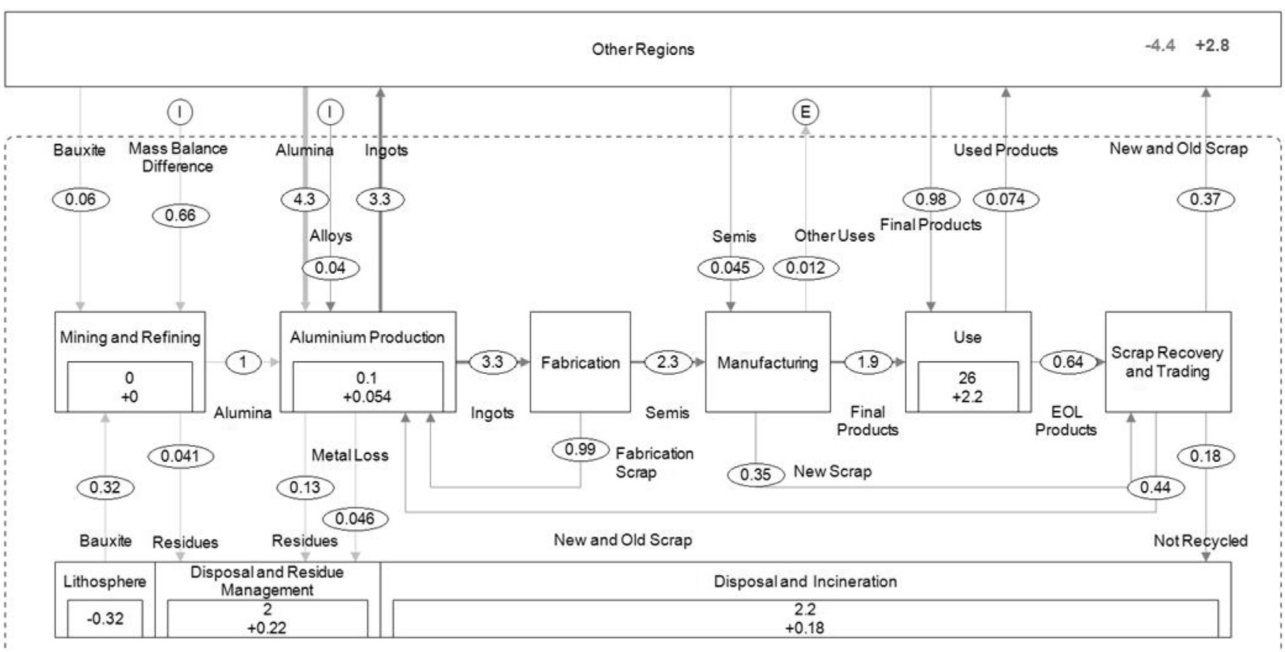
China (Tool), 2014



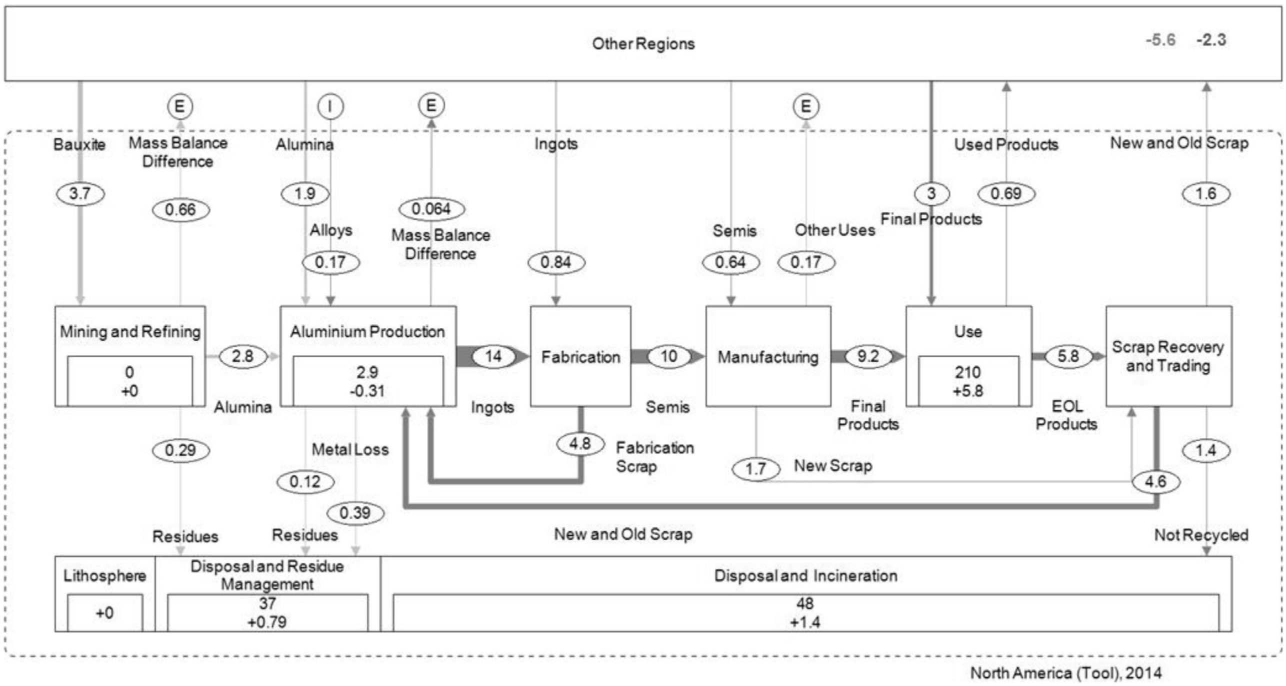
Europe (Tool), 2014

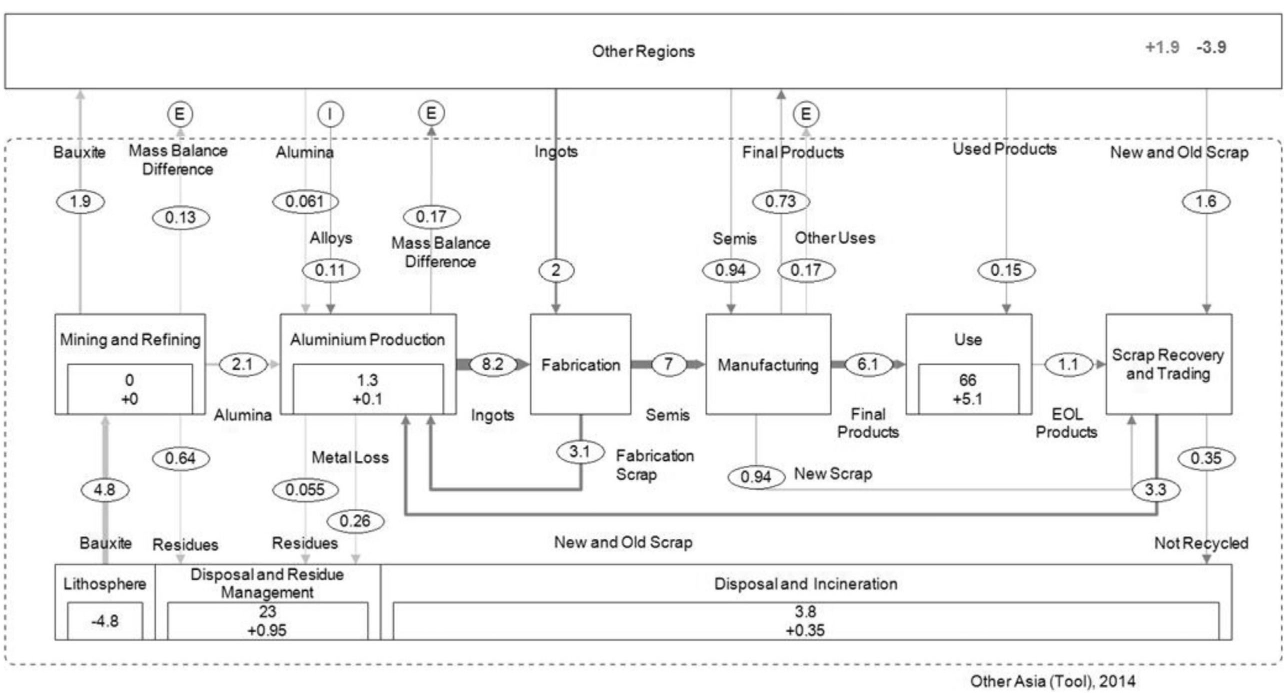


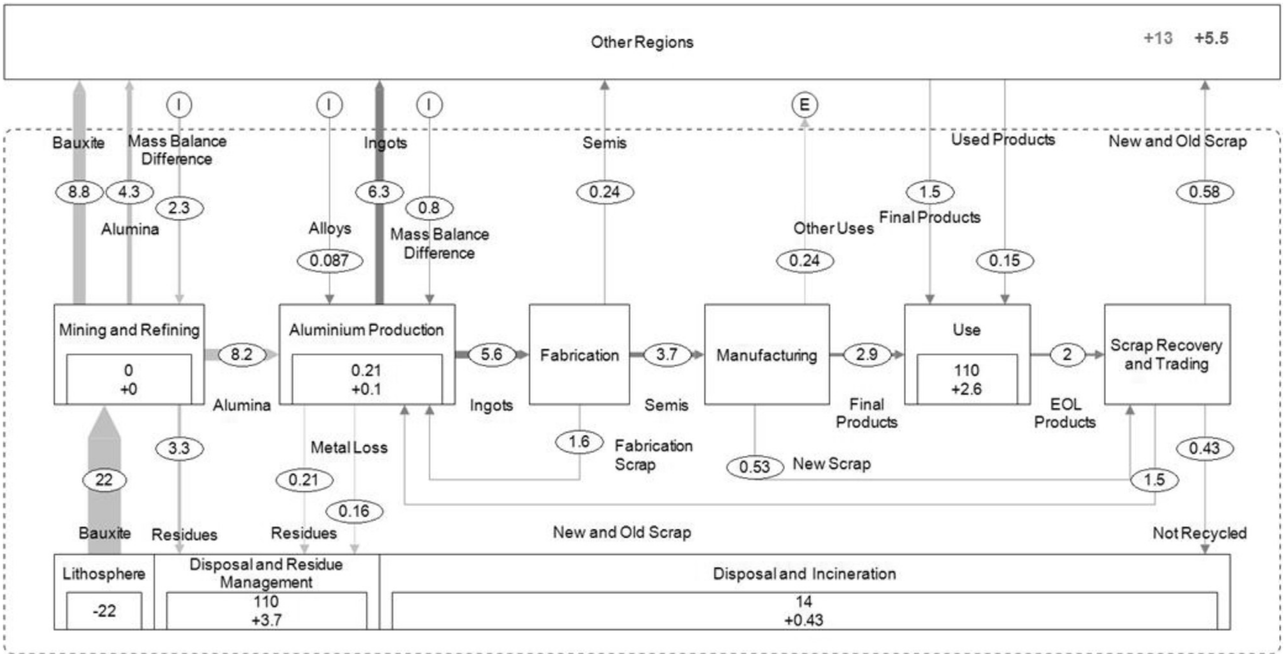
Japan (Tool), 2014



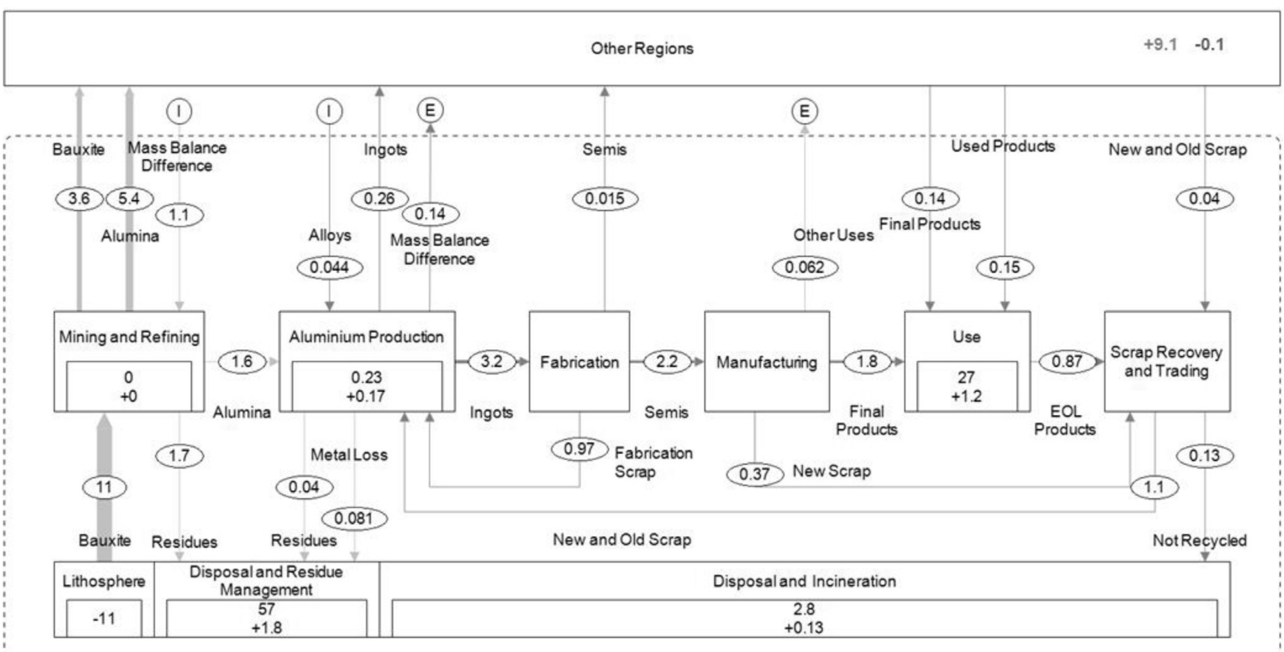
Middle East (Tool), 2014


Other Regions
-5.6 -2.3
Bauxite
3.7
E
Mass Balance Difference
0.66
Alumina
1.9
I
Alloys
0.17
E
0.064
Mass Balance Difference
Ingots
0.84
Semis
0.64
E
Other Uses
0.17
3
Final Products
Used Products
0.69
New and Old Scrap
1.6
Mining and Refining
0
+0
2.8
Alumina
Aluminium Production
2.9
-0.31
14
Ingots
Fabrication
10
Semis
Manufacturing
9.2
Final Products
Use
210
+5.8
5.8
EOL Products
Scrap Recovery and Trading
0.29
Metal Loss
0.12
0.39
4.8
Fabrication Scrap
1.7
New Scrap
4.6
1.4
Residues
Residues
New and Old Scrap
Not Recycled
Lithosphere
+0
Disposal and Residue Management
37
+0.79
Disposal and Incineration
48
+1.4
North America (Tool), 2014

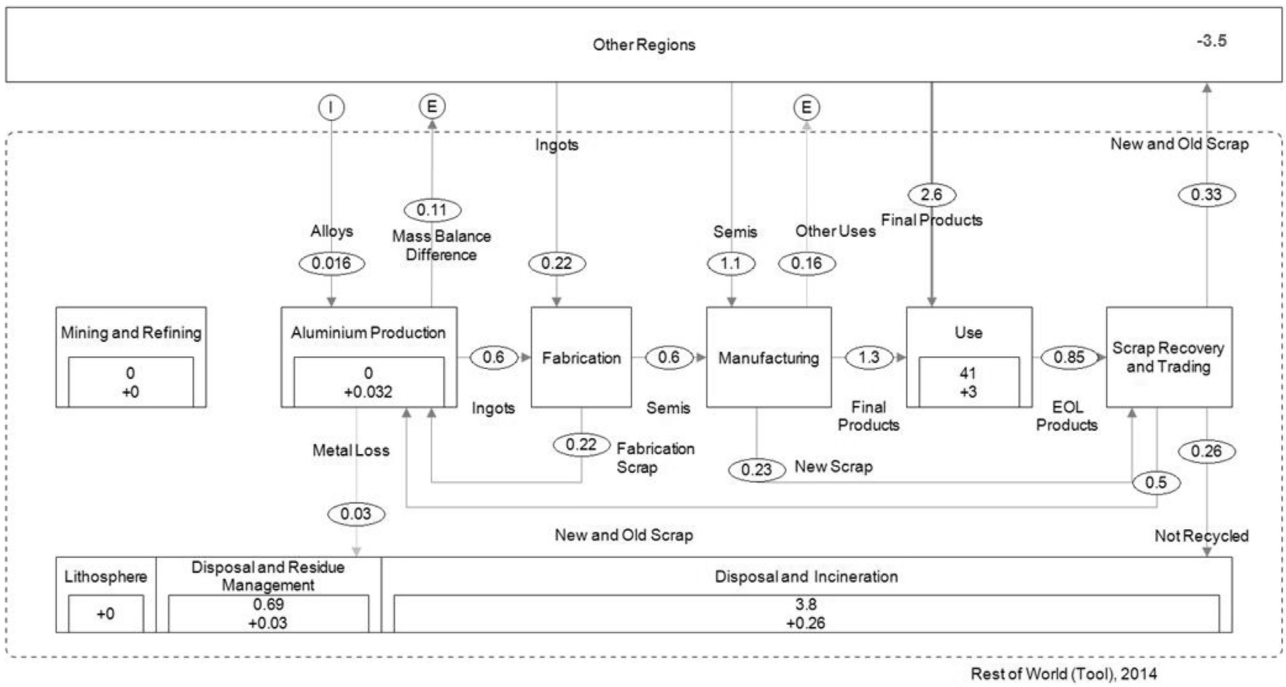

Other Regions
+1.9 -3.9
Bauxite
1.9
E
Mass Balance Difference
0.13
Alumina
0.061
I
Alloys
0.11
E
0.17
Mass Balance Difference
Ingots
2
Final Products
0.73
E
Semis
0.94
Other Uses
0.17
Used Products
0.15
New and Old Scrap
1.6
Mining and Refining
0
+0
2.1
Alumina
Aluminium Production
1.3
+0.1
8.2
Ingots
Fabrication
7
Semis
Manufacturing
6.1
Final Products
Use
66
+5.1
1.1
EOL Products
Scrap Recovery and Trading
0.64
4.8
Bauxite
Metal Loss
0.055
0.26
3.1
Fabrication Scrap
0.94
New Scrap
3.3
0.35
Residues
Residues
New and Old Scrap
Not Recycled
Lithosphere
-4.8
Disposal and Residue Management
23
+0.95
Disposal and Incineration
3.8
+0.35
Other Asia (Tool), 2014



Other Producing Countries (Tool), 2014

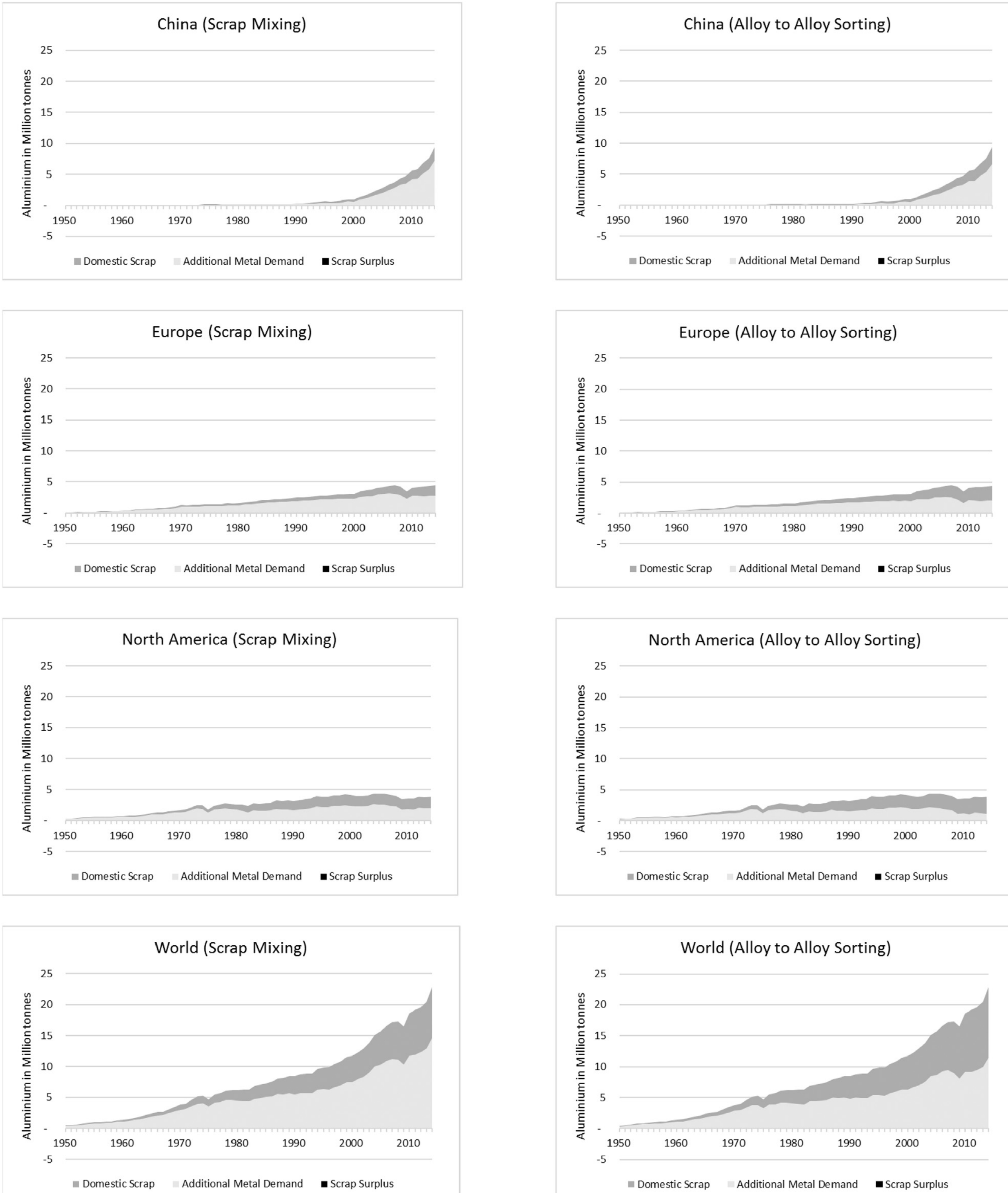


South America (Tool), 2014



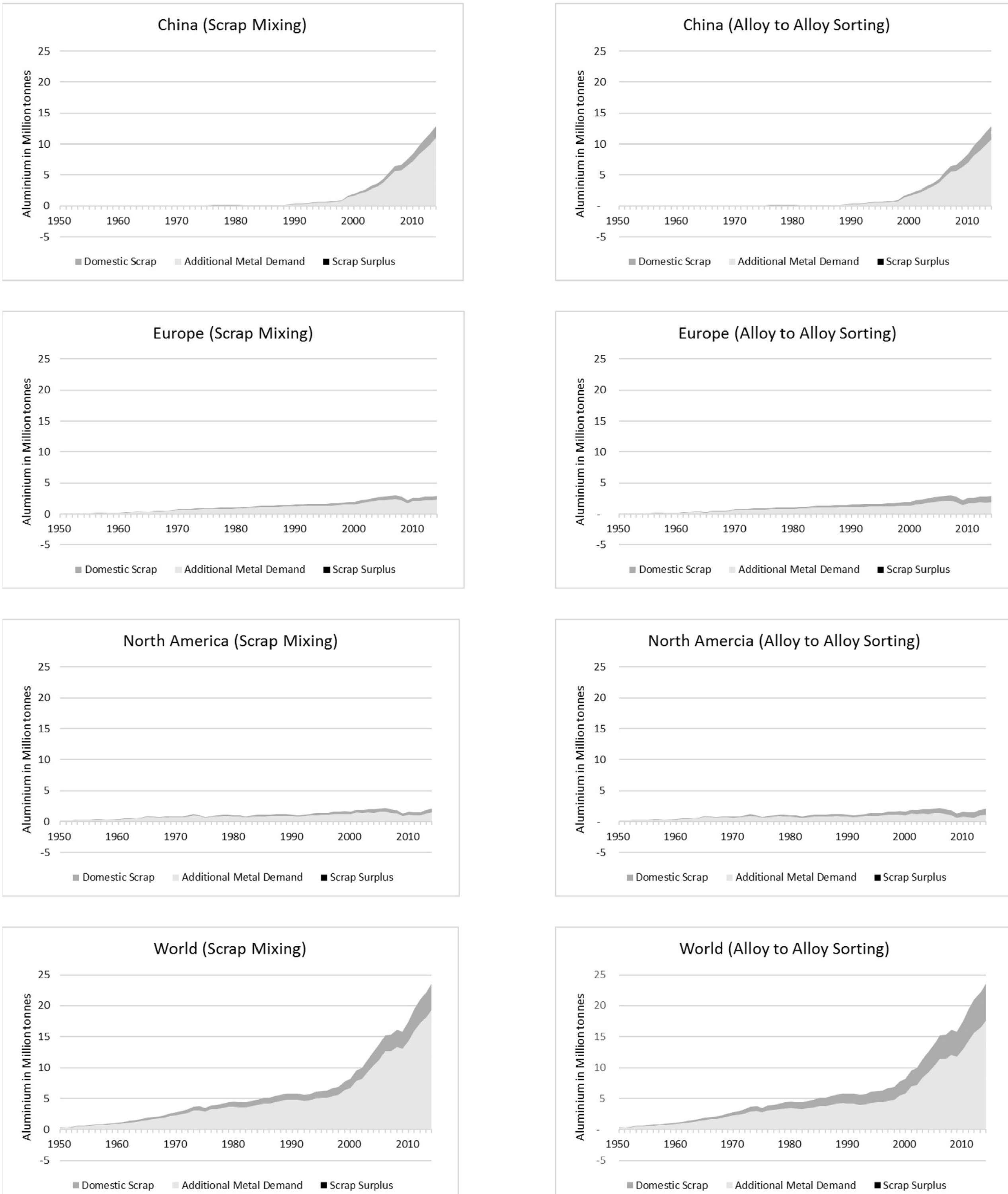
Flows are in million tonnes. Stocks and flows are rounded to two significant digits. Process in- and outputs might not add up due to rounding. Light grey flows represent non-metal and dark grey metal. All flows are converted to aluminium (Bauxite, Alumina, Residues, Metal Loss) and aluminium alloys (Ingots, Semis, Final Products, EOL Products, Used Products and Not Recycled).

Appendix C. Recycled rolled ingot demand and supply balance globally and in selected regions



See Fig. 6 for further information.

Appendix D. Recycled extrusion ingot demand and supply balance globally and in selected regions



See Fig. 6 for further information.

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